1. Prove that OLS estimator is unbiased.

An **estimator** is a rule or formula used to estimate an unknown population parameter based on sample data.
If β is a population parameter, then β^ is its estimator.

Ordinary Least Squares (OLS) is a method used in linear regression to estimate the unknown parameters β by minimizing the sum of squared residuals.

For the linear model:

$$y = X\beta + \varepsilon$$

The OLS estimator is:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Assumptions (Gauss–Markov conditions)

1. Linear model: $y = X\beta + \varepsilon$
2. $E(\varepsilon) = 0$
3. $\mathrm{Var}(\varepsilon) = \sigma^2 I$
4. X has full column rank

Proof that OLS is Unbiased

We want to show:

$$E(\hat{\beta}) = \beta$$

Substitute $y = X\beta + \varepsilon$ into the estimator:

$$\hat{\beta} = (X^T X)^{-1} X^T (X\beta + \varepsilon)$$

$$\hat{\beta} = (X^T X)^{-1} X^T X\beta + (X^T X)^{-1} X^T \varepsilon$$

$$\hat{\beta} = \beta + (X^T X)^{-1} X^T \varepsilon$$

Taking expectation:

$$E(\hat{\beta}) = \beta + (X^T X)^{-1} X^T E(\varepsilon)$$

Since $E(\varepsilon) = 0$,

$$E(\hat{\beta}) = \beta$$

Hence, the OLS estimator is unbiased.

2. Derive covariance of OLS estimator.

Covariance measures how two random variables vary together.

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For two random variables X and Y:

$$\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

If covariance is:

- Positive → variables move in the same direction
- Negative → variables move in opposite directions
- Zero → no linear relationship

For a random vector Z, covariance matrix is:

$$\mathrm{Var}(Z) = E[(Z - E[Z])(Z - E[Z])^T]$$

Consider the linear regression model:

$$y = X\beta + \varepsilon$$

Assumptions:

1. $E(\varepsilon) = 0$
2. $\mathrm{Var}(\varepsilon) = \sigma^2 I$
3. X has full rank

The OLS estimator is:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Substitute $y = X\beta + \varepsilon$:

$$\hat{\beta} = (X^T X)^{-1} X^T (X\beta + \varepsilon)$$

$$\hat{\beta} = \beta + (X^T X)^{-1} X^T \varepsilon$$

Now compute variance (covariance matrix):

$$\mathrm{Var}(\hat{\beta}) = \mathrm{Var}\left(\beta + (X^T X)^{-1} X^T \varepsilon\right)$$

Since β is constant:

$$\mathrm{Var}(\hat{\beta}) = \mathrm{Var}\left((X^T X)^{-1} X^T \varepsilon\right)$$

Use property:

$$\mathrm{Var}(AZ) = A\mathrm{Var}(Z)A^T$$

So,

$$\mathrm{Var}(\hat{\beta}) = (X^T X)^{-1} X^T \mathrm{Var}(\varepsilon) X (X^T X)^{-1}$$

Since $\mathrm{Var}(\varepsilon) = \sigma^2 I$,

$$\mathrm{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1}$$

$$\mathrm{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

Thus, the covariance matrix of the OLS estimator is

$$\sigma^2(X^TX)^{-1}$$

This shows that the variability of OLS depends on:

- Error variance σ^2
- Design matrix X

3. When is OLS BLUE? (Gauss-Markov Theorem)

BLUE stands for Best Linear Unbiased Estimator.

The Ordinary Least Squares (OLS) estimator is BLUE under the Gauss–Markov assumptions.

Gauss–Markov Theorem

The theorem states:

In a linear regression model, if certain assumptions hold, the OLS estimator has the minimum variance among all linear unbiased estimators.

That means OLS is:

- Linear
- Unbiased
- Best (minimum variance)

Conditions for OLS to be BLUE

Consider the linear model:

$$y = X\beta + \varepsilon$$

OLS is BLUE if the following assumptions hold:

1. Linearity in Parameters

The model is linear in β:

$$y = X\beta + \varepsilon$$

2. Zero Mean of Errors

$$E(\varepsilon) = 0$$

Ensures unbiasedness.

3. Homoscedasticity

$$\mathrm{Var}(\varepsilon) = \sigma^2 I$$

Errors have constant variance.

4. No Autocorrelation

Errors are uncorrelated:

$$\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad (i \neq j)$$

5. No Perfect Multicollinearity

X has full column rank ($X^T X$ is invertible).

4. Why does multicollinearity increase variance?

Multicollinearity occurs when two or more independent variables in a regression model are highly correlated. This causes redundancy in the information provided by predictors.

The variance of the OLS estimator is:

$$\mathrm{Var}(\hat{\beta}) = \sigma^2(X^T X)^{-1}$$

- $X^T X$ is the matrix of predictor correlations.
- If predictors are highly correlated, $X^T X$ becomes **nearly singular** (determinant close to 0).
- Inverting a nearly singular matrix leads to **large values in** $(X^T X)^{-1}$.
- Therefore, the variance of $\hat{\beta}$ increases.

5. Show ridge regression shrinks eigenvalues.

In linear algebra, for a square matrix A, a scalar λ is called an eigenvalue if there exists a non-zero vector v (called an eigenvector) such that:

$$Av=\lambda v$$

Ridge Regression is a type of regularized linear regression that adds a penalty on the size of coefficients to reduce overfitting and handle multicollinearity.

- Q = matrix of eigenvectors
- Λ = diagonal matrix of eigenvalues λ_i

Ridge estimator:

$$\hat{\beta}_{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

Eigen-decompose:

$$X^T X = Q \Lambda Q^T$$

Then:

$$X^T X + \lambda I = Q(\Lambda + \lambda I) Q^T$$

Inverse:

$$(X^T X + \lambda I)^{-1} = Q(\Lambda + \lambda I)^{-1} Q^T$$

Each eigenvalue transforms:

$$\lambda_i \to \frac{1}{\lambda_i + \lambda}$$

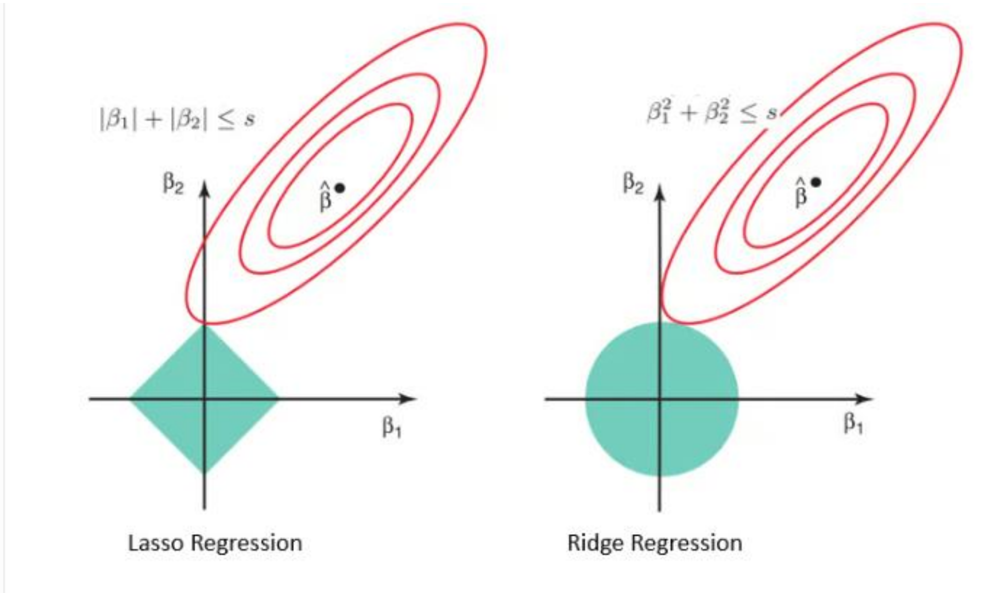
Since $\lambda > 0$,

Denominator increases ⇒ coefficients shrink.

6. Compare ridge vs lasso geometrically.

Characteristic	Ridge Regression	Lasso Regression
Regularization Type	Applies L2 regularization, adding a penalty term proportional to the square of the coefficients	Applies L1 regularization, adding a penalty term proportional to the absolute value of the coefficients.

Feature Selection	Does not perform feature selection. All predictors are retained, although their coefficients are reduced in size to minimize overfitting	Performs automatic feature selection. Less important predictors are completely excluded by setting their coefficients to zero.
When to use	Best suited for situations where all predictors are potentially relevant, and the goal is to reduce overfitting rather than eliminate features	Ideal when you suspect that only a subset of predictors is important, and the model should focus on those while ignoring the irrelevant ones.
Output model	Produces a model that includes all features, but their coefficients are smaller in magnitude to prevent overfitting	Produces a model that is simpler, retaining only the most significant features and ignoring the rest by setting their coefficients to zero.
Impact on Prediction	Reduces the magnitude of coefficients, shrinking them towards zero, but does not set any coefficients exactly to zero. All predictors remain in the model	Shrinks some coefficients to exactly zero, effectively removing their influence from the model. This leads to a simpler model with fewer features
Computation	Generally faster as it doesn’t involve feature selection	May be slower due to the feature selection process
Example Use Case	Use when you have many predictors, all contributing to the outcome (e.g., predicting house prices where all features like size, location, etc., matter)	Use when you believe only some predictors are truly important (e.g., genetic studies where only a few genes out of thousands are relevant).



- In the plot, red ellipses show the contours of the cost function, and the shaded regions show the penalty constraints: diamond for Lasso (L1) and circle for Ridge (L2).
- The regression solution is found where the ellipses first touch the constraint region.
- Lasso's diamond shape has corners on the axes, so intersections often make some coefficients exactly zero, allowing variable selection.
- Ridge's circular shape has no corners, so coefficients are shrunk toward zero but never exactly zero, keeping all variables.
- Relaxing the penalty (larger diamond or circle) moves the solution closer to the OLS solution at the center of the ellipses.
- The key difference is the constraint shape: Lasso → sparsity, Ridge → shrinkage without sparsity.

7. Derive linear regression as MAP estimate.

Maximum a Posteriori (MAP) estimation is a Bayesian technique used to estimate unknown parameters by finding the mode of the posterior distribution, combining observed data (likelihood) with prior knowledge.

We start with the linear regression model:

$$y_i = \mathbf{x}_i^\top \mathbf{w} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

where $\mathbf{x}_i \in \mathbb{R}^d$ is the feature vector for sample i, $\mathbf{w} \in \mathbb{R}^d$ is the weight vector, and ϵ_i is Gaussian noise. The likelihood for a single observation is:

$$p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_i^\top \mathbf{w})^2}{2\sigma^2} \right)$$

For the full dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, assuming independence, the likelihood is:

$$p(\mathbf{y} \mid X, \mathbf{w}) = \prod_{i=1}^n p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \mathbf{w})^2 \right)$$

To perform MAP estimation, we assume a prior on the weights. A common choice is a zero-mean Gaussian prior:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}) = \frac{1}{(2\pi\tau^2)^{d/2}} \exp\left(-\frac{\|\mathbf{w}\|_2^2}{2\tau^2} \right)$$

The MAP estimate maximizes the posterior:

$$\mathbf{w}_{\text{MAP}} = \arg\max_{\mathbf{w}} p(\mathbf{w} \mid X, \mathbf{y}) = \arg\max_{\mathbf{w}} p(\mathbf{y} \mid X, \mathbf{w}) p(\mathbf{w})$$

Taking the log-posterior gives:

$$\log p(\mathbf{w} \mid X, \mathbf{y}) = \log p(\mathbf{y} \mid X, \mathbf{w}) + \log p(\mathbf{w}) + \text{const} = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \mathbf{w})^2 - \frac{1}{2\tau^2} \|\mathbf{w}\|_2^2 + \text{const}$$

Maximizing the log-posterior is equivalent to minimizing its negative, leading to:

$$\mathbf{w}_{\text{MAP}} = \arg\min_{\mathbf{w}} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \mathbf{w})^2 + \lambda \|\mathbf{w}\|_2^2$$

where $\lambda = \frac{\sigma^2}{\tau^2}$. This is exactly ridge regression. If $\tau^2 \to \infty$, then $\lambda \to 0$ and we recover ordinary least squares (OLS).

The closed-form solution for the MAP estimate is:

$$\mathbf{w}_{\text{MAP}} = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$$

where $X \in \mathbb{R}^{n \times d}$ is the design matrix and I is the $d \times d$ identity matrix.

Thus, MAP linear regression naturally introduces L2 regularization through the Gaussian prior, and OLS is a special case when the prior is uninformative.

8. Analyze condition number of XTX.

$$X^T X.$$

The **condition number** is a measure of how sensitive a function or a system of equations is to small changes or errors in its input. In the context of linear algebra and matrices, it tells you how **stable** solving a linear system is.

Formally, for a square matrix A, the **2-norm condition number** is defined as:

$$\kappa(A) = \|A\|_2 \cdot \|A^{-1}\|_2 = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

where:

- $\sigma_{\max}(A)$ is the largest singular value of A,
- $\sigma_{\min}(A)$ is the smallest singular value of A.

The condition number of a matrix is a measure of how sensitive the solution of a linear system is to perturbations in the input. For linear regression, the matrix of interest is $X^\top X$, where $X \in \mathbb{R}^{n\times d}$ is the design matrix. The condition number of $X^\top X$ is defined as:

$$\kappa(X^\top X) = \frac{\sigma_{\max}(X^\top X)}{\sigma_{\min}(X^\top X)}$$

where $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest singular values of $X^\top X$. Since $X^\top X$ is symmetric and positive semi-definite, its singular values are equal to its eigenvalues, so we can equivalently write:

$$\kappa(X^\top X) = \frac{\lambda_{\max}(X^\top X)}{\lambda_{\min}(X^\top X)}$$

A large condition number indicates that $X^\top X$ is **ill-conditioned**, meaning small changes in X or the target vector y can lead to large changes in the regression solution $\mathbf{w} = (X^\top X)^{-1}X^\top y$. Ill-conditioning often occurs when the columns of X are nearly linearly dependent (high multicollinearity).

The condition number is related to the singular values of X itself because $X^\top X$ has eigenvalues equal to the squares of X's singular values:

$$\lambda_i(X^\top X) = \sigma_i(X)^2$$

so the condition number can also be expressed as:

$$\kappa(X^\top X) = \frac{\sigma_{\max}(X)^2}{\sigma_{\min}(X)^2} = \kappa(X)^2$$

This shows that $X^\top X$ is more sensitive to ill-conditioning than X, because the condition number is **squared**. In practice, a high condition number can lead to unstable regression coefficients, large variance, and numerical errors during inversion. Regularization techniques, such as ridge regression, improve stability by adding λI to $X^\top X$, which increases $\lambda_{\min}$ and reduces the effective condition number:

$$\kappa(X^\top X + \lambda I) = \frac{\lambda_{\max}(X^\top X) + \lambda}{\lambda_{\min}(X^\top X) + \lambda}$$

This ensures that the matrix is better conditioned and the solution is numerically more stable.

9. Prove convexity of squared loss.

A function $f : \mathbb{R}^d \to \mathbb{R}$ is **convex** if, for any two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ and any $\theta \in [0,1]$, it satisfies:

$$f(\theta\mathbf{x} + (1-\theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1-\theta) f(\mathbf{y})$$

- Geometrically, this means the **line segment** connecting $f(\mathbf{x})$ and $f(\mathbf{y})$ lies **above or on the graph** of the function.
- If the inequality is strict for $\mathbf{x} \neq \mathbf{y}$, the function is called **strictly convex**.

Consider the squared loss function used in linear regression:

$$L(\mathbf{w}) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \mathbf{w})^2$$

where $\mathbf{w} \in \mathbb{R}^d$ is the weight vector, $\mathbf{x}_i \in \mathbb{R}^d$ is the feature vector, and y_i is the target. To prove convexity, we examine the second derivative (Hessian) with respect to $\mathbf{w}$.

First, rewrite the loss in matrix form:

$$L(\mathbf{w}) = \|\mathbf{y} - X\mathbf{w}\|_2^2 = (\mathbf{y} - X\mathbf{w})^\top (\mathbf{y} - X\mathbf{w})$$

where $X \in \mathbb{R}^{n \times d}$ is the design matrix and $\mathbf{y} \in \mathbb{R}^n$ is the target vector. Expanding the quadratic:

$$L(\mathbf{w}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top X\mathbf{w} + \mathbf{w}^\top X^\top X\mathbf{w}$$

The gradient of $L(\mathbf{w})$ with respect to $\mathbf{w}$ is:

$$\nabla_{\mathbf{w}} L(\mathbf{w}) = -2X^\top \mathbf{y} + 2X^\top X\mathbf{w}$$

The Hessian (second derivative) is:

$$\nabla_{\mathbf{w}}^2 L(\mathbf{w}) = 2X^\top X$$

Since $X^\top X$ is symmetric and positive semi-definite for any real matrix X, it follows that $2X^\top X$ is also positive semi-definite. By definition, a function whose Hessian is positive semi-definite is convex. Therefore, the squared loss $L(\mathbf{w})$ is **convex** with respect to $\mathbf{w}$.

Convexity ensures that any local minimum of the squared loss is also a global minimum, which is why linear regression has a unique solution when $X^\top X$ is invertible.

10. What happens when p >> n?

When $p \gg np$ (i.e., the number of features p is much larger than the number of samples n), we enter the **high-dimensional**

regime, and this has significant implications for linear regression.

When $p \gg n$:

- $X^\top X$ is **singular** and not invertible.
- OLS has **infinitely many solutions** and overfits the data.
- Regression coefficients are **unstable** (high condition number).
- **Regularization** (Ridge/Lasso) or **dimensionality reduction** is necessary to get stable, meaningful estimates.